

Research on Energy System Management and Circuit Optimization Strategies for Unmanned Underwater Vehicles

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Abstract—With the continuous development of UUV (Unmanned Underwater Vehicles) equipped with acoustic devices towards greater depth and longer range, the requirements for internal energy and circuit design of the vehicles are increasing; The purpose of this paper is to study and optimize the mainstream lithium-ion battery system and load power circuit of UUV, achieve accurate system state feedback, reduce instantaneous power demand and energy system design redundancy. The specific method is to build an experimental platform to conduct charge and discharge tests on two groups of batteries, and obtain characteristic data at different aging stages; Establish a second-order equivalent circuit model, use the recursive least squares algorithm to identify model parameters online, design a state estimation model based on extended Kalman filtering, and correct the model estimation accuracy based on the current capacity state of the battery mapped by the capacity increment representation; Design efficient matching circuits based on the characteristics of load impedance. The results show that the proposed method has high accuracy in state feedback and significant circuit optimization effect, which can provide effective reference for the design of power supply and distribution systems for UUV.

Keywords—unmanned underwater vehicle, Li-ion battery, energy system, circuit optimization

I. INTRODUCTION

The ocean is a huge treasure trove of resources, and human development of the ocean is still shallow. Taking marine oil and gas development as an example, countries around the world are currently extending their development to the deep sea, and the transfer of oil and gas development systems from the surface to underwater is also an inevitable trend. UUV for underwater development operations are currently the main form of industry development and an indispensable technical means for various industries to carry out underwater exploration and marine resource development[1]. In recent years, there have been a number of UUV with diving depths of hundreds of kilometers or even tens of thousands of meters in the world[2]. The greater the diving depth of the UUV, the longer the operation time, and the higher the requirements for the energy system. In addition to basic navigation and propulsion equipment, most UUV are

equipped with acoustic loads that are essential for active underwater detection to meet the requirements of underwater operations. They are also one of the main equipment that requires platform power. How to improve the energy utilization efficiency of load equipment and reduce the power demand on UUV is also urgently needed for research.

The energy system development of UUV mainly includes lead-acid batteries, zinc silver batteries, lithium-ion batteries, and fuel cells. Lead acid batteries have low specific energy, short cycle life, and poor reliability, while zinc silver batteries have long charging time, poor low-temperature performance, and high cost, which cannot meet the operational requirements of current UUV; The complexity and poor safety of fuel cell systems also limit their application in UUV; Lithium-ion batteries, with their outstanding performance advantages such as no memory effect, high specific energy, and low self-discharge rate[3-4], have been well applied in UUV.

The energy system of UUV is limited by spatial structure size and weight, and generally uses high-energy density lithium-ion batteries to meet the requirements of endurance and operation. Lithium-ion battery systems are generally composed of multiple individual battery cells, and there is inconsistency between the batteries. In order to improve energy utilization efficiency, precise feedback and control of the entire battery system are required, which places high demands on battery state estimation; In addition, UUV need to be designed and allocated according to the maximum power requirements of the load they carry. To reduce the apparent power of the system, it is important to design efficient impedance matching circuits. This article focuses on the research and design of feedback control strategies for energy systems applicable to UUV, while optimizing circuits based on load characteristics to reduce the power design capacity of UUV and improve the reliability of energy system control.

II. ENERGY SYSTEM

Lithium-ion batteries, as the mainstream energy source for UUV, are crucial for precise state monitoring and feedback in

platform internal control. This article takes the commonly used ternary NCM lithium-ion battery as the object and validate the algorithm through measured data.

A. Battery Experiment

This article uses ternary nickel cobalt manganese batteries as experimental batteries, as shown in Figure 1:



Fig. 1. Experimental battery.

Conduct charge and discharge tests on two sets of batteries to obtain battery data under different cycling states. To analyze the internal resistance characteristics of the battery, hybrid pulse power characteristic test was conducted[5]. The process involved fully charging the battery, applying discharge and charge pulse currents to the battery, and recording the changes in battery voltage. The test results are shown in Figure 2:

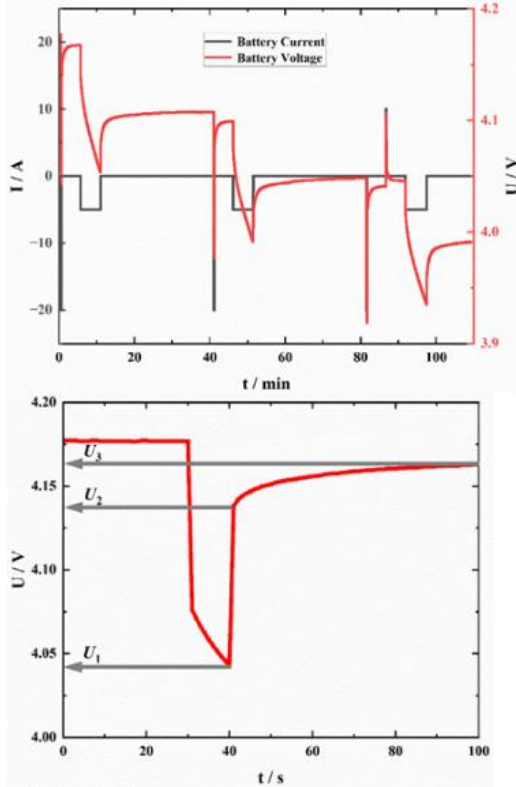


Fig. 2. Hybrid pulse power characteristic test.

Where I is the value of pulse discharge current; U_1 , U_2 , U_3 are the initial voltage value after the discharge pulse ends, the voltage value at the first sampling point after the pulse ends, and the voltage value after the voltage stabilizes. The internal impedance information of the battery can be obtained offline through the collected external characteristic parameters.

To better simulate the operating conditions that may be encountered during actual operation, dynamic operating condition tests were conducted on the battery system, recording changes in battery current and voltage. The relevant results are shown in Figure 3:

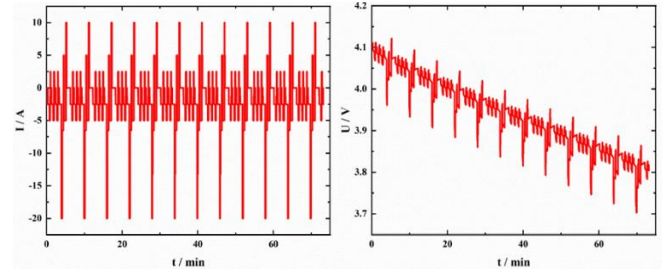


Fig. 3. Battery dynamic condition testing.

Verify the accuracy of the system algorithm based on the obtained external characteristic data under dynamic operating conditions.

The voltage difference measured when the positive and negative electrodes of a lithium-ion battery are disconnected is the open circuit voltage of the battery. The degree of aging, temperature, and state of charge of the battery will all affect the open circuit voltage. Among them, there is a clear mapping relationship between the open circuit voltage and the state of charge of the battery at the same degree of aging and temperature. A model can be established to characterize the nonlinear functional relationship between the open circuit voltage and the state of charge. Each type of power battery has its basic stable OCV-SOC curve[6], and as aging intensifies, the open circuit voltage characteristics of the battery will undergo significant changes. Accurate calibration of this curve is of great significance for the design of algorithm strategies such as lithium-ion battery modeling and battery state estimation. This article conducts open circuit voltage testing on battery systems, records OCV-SOC changes, and fits the results using a 7th order polynomial.

$$U_{oc} = p_0 + p_1 * SOC + p_2 * SOC^2 + p_3 * SOC^3 + p_4 * SOC^4 + p_5 * SOC^5 + p_6 * SOC^6 + p_7 * SOC^7 \quad (1)$$

The relevant results are shown in Figure 4:

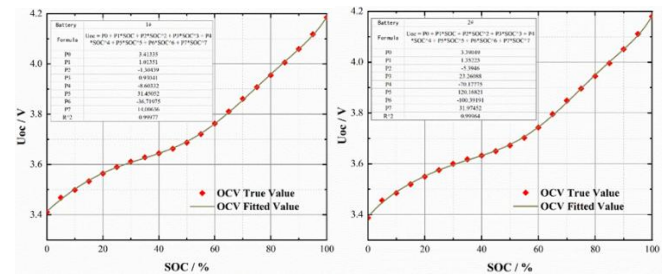


Fig. 4. OCV-SOC curve of experimental battery.

B. Battery Modeling

The equivalent circuit model with RC network as the core can well describe the dynamic external characteristics of the battery, including Ohmic characteristics and polarization characteristics. Its model has advantages in both accuracy and computational complexity compared to other models. Given the precision and real-time requirements of UUV for lithium-ion battery management system in battery state estimation, first-order models cannot reflect the different polarization effects of batteries, while third-order and above RC network models require a large amount of computation, which will slow down the processing speed of the controller and result in significant errors in the calculation results. Therefore, this article uses a second-order RC equivalent circuit model to analyze the battery system[7], as shown in Figure 5.

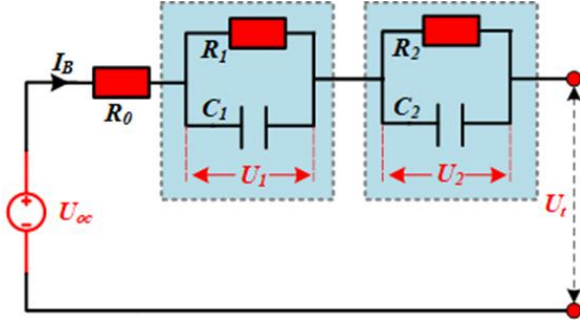


Fig. 5. Second-order equivalent circuit model.

Among them, U_{oc} represents the open circuit voltage of the battery, U_t represents the terminal voltage, I_B represents the main current, R_0 represents the ohmic resistance, R_1 and R_2 represent the polarization resistance, C_1 and C_2 represent the polarization capacitance, U_1 and U_2 represent the polarization voltage.

$$U_{oc} = U_t + \sum_{i=1}^2 U_i + I_B R_0 \quad (2)$$

$$\dot{U}_i = \frac{I_B}{C_i} - \frac{U_i}{R_i C_i}; (i=1, 2) \quad (3)$$

Due to the short sampling time, it can be assumed that the equivalent circuit model parameters of the lithium-ion battery remain unchanged during this period. To simplify the calculation of the time-varying steady-state system, the following equation can be obtained.

$$U_i[(k+1)\Delta t] = e^{\frac{-1}{R_i C_i} \Delta t} U_i[k\Delta t] + \int_{k\Delta t}^{(k+1)\Delta t} e^{\frac{-1}{R_i C_i} [(k+1)\Delta t - \tau]} \frac{1}{C_i} I_B(\tau) d\tau \quad (4)$$

Among them, τ is a time constant, and during the discretization process period, the battery current can be regarded as constant per unit time. Defining the subscript k as time k , we can obtain:

$$U_{i,k+1} = e^{\frac{-1}{R_i C_i} \Delta t} U_{i,k} + R_i I_{B,k+1} \left[1 - e^{\frac{-1}{R_i C_i} \Delta t} \right] \quad (5)$$

According to the ampere time integral, we can obtain:

$$SOC_{k+1} = SOC_k - I_{B,k+1} \frac{\Delta t}{C_n} \quad (6)$$

Among them, C_n is the battery capacity, and the state space description for establishing the equivalent circuit model of lithium-ion batteries is:

$$\begin{bmatrix} U_{1,k+1} \\ U_{2,k+1} \\ SOC_{k+1} \end{bmatrix} = \begin{bmatrix} \exp(-\Delta t / R_1 C_1) & 0 & 0 \\ 0 & \exp(-\Delta t / R_2 C_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U_{1,k} \\ U_{2,k} \\ SOC_k \end{bmatrix} + \begin{bmatrix} R_1 (1 - \exp(-\Delta t / R_1 C_1)) \\ R_2 (1 - \exp(-\Delta t / R_2 C_2)) \\ \Delta t / C_n \end{bmatrix} \times I_B \quad (7)$$

$$U_{t,k} = U_{oc}(SOC_k) - \sum_{i=1}^2 U_{i,k} - I_{B,k} R_0 \quad (8)$$

C. Parameter Identification

In order to improve the reliability and scalability of the second-order battery model, it is necessary to accurately identify the parameters of the battery model. Offline identification of parameters through the current voltage response curve of battery HPPC can obtain the initial values of the battery model. In this paper, the recursive least squares method is used to identify the battery model parameters online at each moment, and the results can effectively improve the accuracy of the model. Before conducting least squares parameter identification, it is necessary to convert the equivalent circuit model into an ARMA model, which includes:

$$U_{oc}(s) = U_t(s) + I_B(s) \left(R_0 + \sum_{i=1}^2 \frac{R_i}{1 + s R_i C_i} \right) \quad (9)$$

The transfer function of the above equation is:

$$G(s) = \frac{U_{oc}(s) - U_t(s)}{I_B(s)} = \frac{R_0 \tau_2 + R_2 \tau_1 + \sum_{i=1}^2 R_0 \tau_i}{R_0 s^2 + \frac{\tau_1 \tau_2}{s} + \frac{R_0 + \sum_{i=1}^2 R_i}{\tau_1 \tau_2} s + \frac{1}{\tau_1 \tau_2}} \quad (10)$$

In the above equation, $G(s)$ is the circuit impedance, $\tau_i = R_i C_i$ is the time constant, and in the discrete time domain:

$$U_{OC,k} - U_{t,k} = [I_{B,k} \quad I_{B,k-1} \quad I_{B,k-2} \quad U_{t,k-1} - U_{OC,k-1} \quad U_{t,k-2} - U_{OC,k-2}]^T * M \quad (11)$$

$$M = [\beta_0 \quad \beta_1 \quad \beta_2 \quad -\alpha_1 \quad -\alpha_2] \quad (12)$$

The above formula can optimize the M-array parameters through the recursive least squares algorithm. By simplifying and comparing, the model parameters can be obtained as follows:

$$\left\{ \begin{array}{l} R_0 = \frac{\beta_0 - \beta_1 + \beta_2}{1 - \alpha_1 + \alpha_2} \\ R_1 = \left(\tau_1 \frac{\beta_0 + \beta_1 + \beta_2}{1 + \alpha_1 + \alpha_2} + \tau_2 R_0 - \frac{\beta_0 + \beta_1 + \beta_2}{1 + \alpha_1 + \alpha_2} \right) / (\tau_1 - \tau_2) \\ R_2 = \frac{\beta_0 + \beta_1 + \beta_2}{1 + \alpha_1 + \alpha_2} - R_1 - R_0 \\ C_1 = \tau_1 / R_1 \\ C_2 = \tau_2 / R_2 \end{array} \right. \quad (13)$$

Based on the initial parameters determined by HPPC testing, complete online parameter identification.

III. ENERGY SYSTEM STATUS MONITORING

From the state space equation of the aforementioned battery model, the state vector x of the lithium-ion battery can be obtained:

$$x = \begin{bmatrix} U_1 \\ U_2 \\ SOC \end{bmatrix} \quad (14)$$

In response to the nonlinear changes and initial state accuracy issues within the energy system, this paper adopts the adaptive extended Kalman filter algorithm to estimate the state based on real-time collected voltage, current, and temperature data. Assuming the covariance matrix is P , the system noise matrix is Q , the observed noise is R , and the window size is J , the process is as follows:

Parameter initialization: state vector x_0 , covariance matrix P_0 , system noise Q_0 , observation noise R_0 ; According to the discrete expression and parameter identification results of the equivalent circuit model, the state matrix A , input matrix B , and output matrix C of the state space model can be obtained as follows:

$$\left\{ \begin{array}{l} A_{k+1} = \begin{bmatrix} \exp(-\Delta t / R_{1,k+1} C_{1,k+1}) & 0 & 0 \\ 0 & \exp(-\Delta t / R_{2,k+1} C_{2,k+1}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ B_{k+1} = \begin{bmatrix} R_1 (1 - \exp(-\Delta t / R_{1,k+1} C_{1,k+1})) \\ R_2 (1 - \exp(-\Delta t / R_{2,k+1} C_{2,k+1})) \\ \Delta t / C_n \end{bmatrix} \\ C_{k+1} = \begin{bmatrix} \frac{dU_{oc}(SOC_k)}{dSOC_k} & -1 & -1 \end{bmatrix} \end{array} \right. \quad (15)$$

Prior estimation, updating the state vector and covariance matrix over time:

$$\hat{x}_{k+1}^- = A_{k+1} \hat{x}_k^+ + B_{k+1} I_{B,k+1} \quad (16)$$

$$P_{k+1}^- = A_{k+1} P_k^+ A_{k+1}^T + Q_k \quad (17)$$

Error update:

$$\hat{U}_{t,k+1} = U_{oc} \left(soc_{k+1} \right) - \sum_{i=1}^2 \hat{U}_{i,k+1}^- - I_{B,k+1} R_{0,k+1} \quad (18)$$

$$e_{k+1} = U_{t,k+1} - \hat{U}_{t,k+1} \quad (19)$$

Gain matrix:

$$K_{k+1} = P_{k+1}^- C_{k+1}^T (C_{k+1} P_{k+1}^- C_{k+1}^T + R)^{-1} \quad (20)$$

Adaptive noise covariance matching:

$$H_{k+1} = \frac{1}{J} \sum_{i=k-J}^{k+1} e_i e_i^T \quad (21)$$

$$R_{k+1} = H_{k+1} - C_{k+1} P_{k+1}^- C_{k+1}^T \quad (22)$$

$$Q_{k+1} = K_{k+1} H_{k+1} K_{k+1}^T \quad (23)$$

Status update:

$$\hat{x}_{k+1}^+ = \hat{x}_{k+1}^- + K_{k+1} e_{k+1} \quad (24)$$

Covariance update:

$$P_{k+1}^+ = (E - K_{k+1} C_{k+1}) P_{k+1}^- \quad (25)$$

Let $k=k+1$ and repeat the above steps.

In the above loop, " $\hat{\cdot}$ " represents the estimated value obtained from the state equation, " $-$ " represents the prior estimate, and " $+$ " represents the posterior estimate.

From the state estimation process, it can be seen that the OCV-SOC mapping relationship is a key point in state estimation. If the OCV-SOC mapping relationship is not accurate, it will lead to significant errors in the estimation results. The degradation of battery capacity is the main factor affecting the OCV-SOC relationship. To know the accurate OCV-SOC mapping relationship, it is necessary to first understand the degradation of battery capacity. Battery capacity is an inherent characteristic of a battery and cannot be directly characterized by external characteristic parameters, so it is necessary to find highly correlated characterization factors. The capacity and voltage under different cycling states have corresponding relationships[8]. Based on experimental data and aging experimental data in this paper, indirect characterization was explored using two sets of battery experimental data. The results are shown in Figure 6.

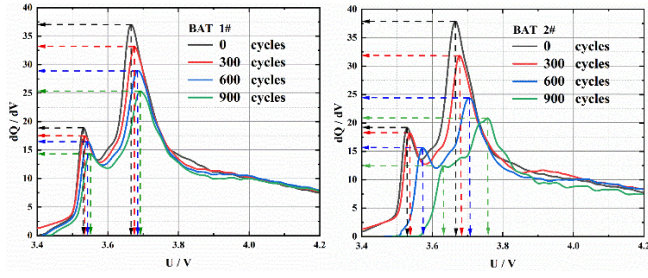


Fig. 6. Trend curve of battery capacity increment.

From the above figure, it can be seen that as the number of cycles increases, the battery capacity increment curve shifts significantly. Based on this, the current capacity state of the battery can be characterized, and the system state estimation accuracy can be corrected.

IV. LOAD CIRCUIT OPTIMIZATION

There are many internal loads in UUV, and some loads (such as transducers) are not purely resistive, and the impedance will change with the working conditions. Therefore, it is necessary to design a reasonable impedance matching circuit to optimize the circuit. This article combines the prior load impedance variation characteristics to design an adaptive impedance matching circuit, ensuring that the load maintains a low peak power demand under different operating conditions. The process is shown in Figure 7.

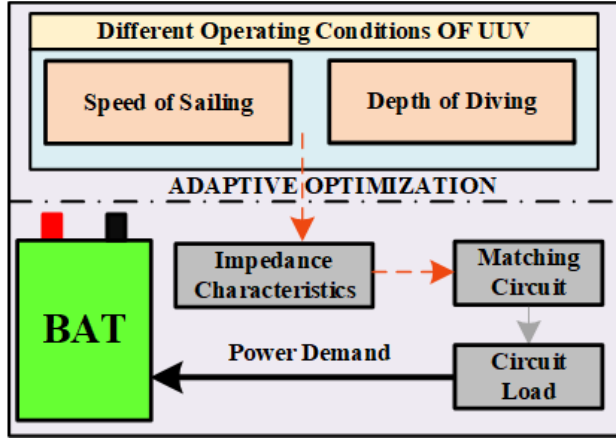


Fig. 7. Optimization process of load circuit.

As shown in the figure, the load circuit is adaptively optimized according to the operating conditions of the unmanned platform, ensuring a high power factor of the circuit and reducing the apparent power demand of the load, thereby reducing the burden on the energy storage system.

V. RESULT VERIFICATION AND ANALYSIS

Based on the dynamic test data of two sets of experimental batteries, this paper validates the proposed state estimation algorithm. The actual initial state of charge is 60%, and the preset initial states of charge are 90% and 20%, respectively. The estimation results are shown in Figure 8:

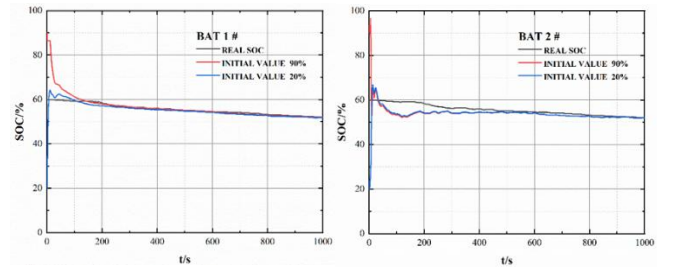


Fig. 8. Battery state estimation results.

The estimation results from two sets of experimental battery data show that the proposed algorithm can converge well to the true state value when the initial preset state is 90% and 20%. The estimated average absolute percentage errors of the first group of batteries are 1.68% and 1.46%, respectively; The estimated average absolute percentage errors of the second group of batteries are 3.66% and 4.26%, respectively.

Based on the load impedance test data of three unmanned platform operating conditions, this paper validates the effectiveness of the proposed circuit optimization. The maximum output current at different frequency points before and after optimization is compared while ensuring a constant output voltage. The results are shown in Figure 9:

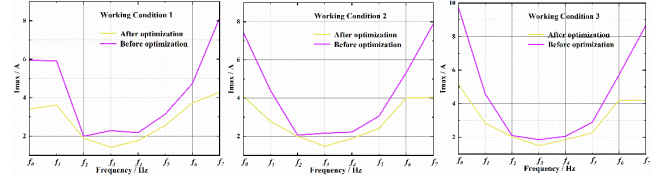


Fig. 9. Circuit optimization effect.

From the circuit optimization results under three operating conditions, it can be seen that the proposed circuit optimization strategy can effectively reduce the maximum output current requirement of the load. The maximum output current of the load circuit for the three operating conditions within the working frequency band decreased by 47.5%, 47.93%, and 47.01% respectively after circuit optimization.

VI. CONCLUSION

(1) The proposed state management method for the energy system of UUV has good estimation performance under dynamic testing conditions. Both sets of experimental batteries can converge to near the true value through the proposed algorithm when the initial preset state deviates from the true state; The estimated average absolute percentage errors of the two battery groups are 1.68%, 1.46%, and 3.66%, 4.26%, respectively, which can accurately estimate the true state of the energy system;

(2) The proposed circuit optimization strategy for UUV has good optimization effects under different operating conditions, and the maximum output current of the load circuit within the operating frequency band is significantly reduced; The maximum output current of the load circuit under three operating conditions decreased by 47.5%, 47.93%, and 47.01%, respectively, reducing the circuit size and lowering the apparent power demand on the energy system.

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